



04/12/2017

SPACE METRICS Inc.
1875 I Street NW, Suite 500
Washington, DC. 20006

Attention: Samuel N. Trotman

Subject: IDIQ Letter Subcontract #2017-00282

Reference: Program Name: Enterprise Programs, Infrastructure, and Communications Services
Noblis Prime Contract Number: DTFAWA-17-D-00014

Effective Date: 4/3/2017

Dear Mr. Trotman:

Noblis, Inc. (Noblis) issues this Letter Subcontract to SPACE METRICS Inc. ("Subcontractor") in anticipation of awarding an IDIQ subcontract, to be issued under Noblis's Prime Contract Number DTFAWA-17-D-00014, in order for the Subcontractor to commence immediate performance of work is further described in the Statement of Work incorporated herein as Attachment A.

SECTION I: GENERAL

By accepting this Letter Subcontract, the Subcontractor agrees to commence negotiations promptly and in good faith with Noblis for the purpose of executing on or before the date specified in Section IV, below, a definitized IDIQ subcontract for the work, for which a certain amount is hereby funded in accordance with Section II, Letter Subcontract Funding.

SECTION II: FUNDING LIMITATION

Noblis's maximum liability under this Letter Subcontract, including termination as provided for in Section V. herein, is limited to total costs in the amount of \$8,000.00 provided that Noblis may, at its sole discretion, by written notice to Subcontractor increase such maximum liability amount. Subcontractor shall provide a minimum of (15) days advance notice of the anticipated commitment or expenditure of 75% of the authorized funding hereunder. Subcontractor shall not be obligated to incur costs or expend funds in excess of the amount stated above.

SECTION III: PERFORMANCE PERIOD

The effective date of this Letter Subcontract is **4/3/2017**. This Letter Subcontract shall expire on **6/20/2017** unless (a) Noblis by written notice to Subcontractor extends the above specified expiration date; or (b) Noblis terminates this Letter Subcontract in accordance with one of the provisions specified in Section V., below. Subcontractor shall not be obligated to perform services beyond the above specified expiration date.

SECTION IV: DEFINITIZATION SCHEDULE

Subcontractor Proposal Submitted	3/20/2017
Negotiations Initiated	4/10/2017
Target Date to Definitize	6/20/2017

SECTION V: TERMINATION

This Letter Subcontract will become part of the definitized subcontract between Noblis and Subcontractor which shall supersede the provisions of this Letter Subcontract. In addition Noblis may terminate this Letter Subcontract, by written notice, for any of the following causes:

- Failure of Noblis and Subcontractor to negotiate and execute a definitized subcontract within the time period specified in Section 4 above.
- Noblis's issuance of a termination notice, for convenience or default

SECTION VI: SUBCONTRACTS TERMS AND CONDITIONS

All certifications and representations which Subcontractor has submitted to Noblis in connection with the eventual award of the subcontract are incorporated herein and made a part hereof and such have been relied upon by Noblis in issuing this Letter Subcontract. Subcontractor agrees to promptly advise Noblis should there be any change in Subcontractor's status with respect to the matters covered by such certifications and representations.

The draft subcontract terms and conditions will be provided shortly and include any such other terms and conditions as may be required by the Prime Contract and/or law or regulation, and those upon which Noblis and Subcontractor may mutually agree are included as terms to the letter subcontract.

SECTION VII: RATED ORDER

Work to be accomplished hereunder is assigned a priority rating of N/A under the referenced Government Prime Contract.

"This is not a rated order certified for national defense use, and you are not required to follow all the provisions of the Defense Priorities and Allocations System regulation (15 CFR 700)."

SECTION VIII: GRATUITIES OR KICKBACKS

- a) No gratuities (in the form of entertainment, gifts or otherwise) or kickbacks shall be offered or given by the Subcontractor with a view toward securing favorable treatment as a supplier.
- b) By accepting this Subcontract, the Subcontractor certifies and represents that it has not made or solicited and will not make or solicit kickbacks in violation of the Anti-Kickback Act of 1986 (41 USC 51-58).
- c) Anti-Kickback Act of 1986
Anti-Kickback Act of 1986 as referenced in FAR 52.203-7 is hereby incorporated into this Request for Proposal as a condition of acceptance. If you have reasonable grounds to believe that a violation, as described in Paragraph (b) of FAR 52.203-7 may have occurred, you should report this suspected

violation to the Noblis's Ethics Hotline at: 888.297.4192 or the Noblis, Inc. Compliance Hotline site for web-based reports: www.ethicspoint.com. You may report a suspected violation anonymously.

SECTION IX: SUBCONTRACTOR'S ACCEPTANCE OF LETTER SUBCONTRACT

Subcontractor's written acknowledgement to Noblis of receipt and acceptance of this Letter Subcontract is required no later than **COB 4/13/2017**.

SECTION X: SUBCONTRACTOR CERTIFICATIONS

By signing this letter subcontract and commencement of work, SPACE METRICS Inc., hereby certifies that neither it nor its principals are presently debarred, suspended, proposed for disbarment, declared ineligible, or voluntarily excluded from participation in this transaction by any Federal department or agency.

By signing this letter subcontract and commencement of work, SPACE METRICS Inc., hereby certifies to the best of its knowledge and belief that no Federal appropriated funds have been paid or will be paid to any person for influencing or attempting to influence an officer or employee of any agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress on its behalf in connection with the awarding of this letter subcontract.

Should you require any additional information or clarifications please contact John L Johnson, Lead Subcontracts Administrator, at (202) 863-3645 or by email at John.johnson@noblis.org.

SPACE METRICS Inc.

Signature:

Printed Name:

Title:


SAMUEL TROTSMAN
PRESIDENT

Noblis, Inc.

Signature:

Printed Name:

Title:

John L
Johnso
n

Digitally signed by
John L Johnson
DN: cn=John L
Johnson, o=Noblis, ou,
email=john.johnson@
noblis.org, c=US
Date: 2017.04.14
10:53:56 -04'00'

Attachment A
STATEMENT OF WORK

Table of Contents

C.1	INTRODUCTION	6
C.1.1	Objective.....	6
C.1.2	Background.....	6
C.1.3	Scope	6
C.2	GENERAL	8
C.2.1	Terminology	8
C.2.2	Applicable Documents.....	8
C.2.3	Government-Furnished Property (GFP).....	8
C.2.3.1	Wireless Internet Access and Remote Access to the FAA Intranet.....	9
C.2.3.2	Workspace, Telephones, and Computers at Governmental Sites	9
C.2.3.3	Other GFP	9
C.2.4	Security Requirements	9
C.2.5	Data Deliverables.....	9
C.2.6	Quality Management.....	9
C.3	FUNCTIONAL TASK AREAS	10
C.3.1	System Engineering	10
C.3.1.1	Requirements Definition and Analysis.....	10
C.3.1.2	Concept Analysis.....	11
C.3.1.3	Enterprise Architecture Planning and Analysis.....	11
C.3.1.4	Standards Development, Validation, and Coordination	12
C.3.1.5	Program- and Project-level Engineering	13
C.3.1.6	System Development and Integration.....	14
C.3.1.7	Test and Evaluation Support	15
C.3.1.8	System Production.....	15
C.3.1.9	System Performance.....	16
C.3.1.10	Interface Management.....	16
C.3.1.11	Information Security	16
C.3.1.12	Governance	17
C.3.1.13	Flight Procedures	18
C.3.1.14	Human Factors	18
C.3.2	Acquisition and Program-management Support	19
C.3.2.1	Acquisition Planning and Support.....	19
C.3.2.2	Program-management Support.....	20
C.3.2.3	Outreach	20
C.3.2.4	Solution-development and Implementation Support	21
C.3.2.5	Program Management Assessments and Reporting	21
C.3.2.6	Quality Assurance	22
C.3.2.7	Risks, Issues, and Opportunities (RIO) Management.....	22
C.3.2.8	Organizational Metrics	23
C.3.2.9	Configuration Management.....	23
C.3.3	In-Service Management and Support.....	24
C.3.3.1	Planning and Coordination.....	24
C.3.3.2	In-Service Management	25
C.3.3.3	Sustainment	26
C.3.3.4	Logistics Support.....	26

C.3.3.5	Performance Assessment and Reporting	27
C.3.3.6	Operational Improvements and Field-Operations Support.....	27
C.3.3.7	Product and Integration Support.....	28
C.3.4	Business and Financial-Management Support	29
C.3.4.1	Budget Planning and Formulation.....	29
C.3.4.2	Budget Execution and Tracking	30
C.3.4.3	Investment and Business-Case Analyses.....	31
C.3.4.4	Cost Estimating	31
C.3.5	Information-Systems Development and Support	31
C.3.5.1	Enterprise Management Tools Development and Support	32
C.3.5.2	Helpdesk Support	34
C.3.5.3	Common Platform Support	34
C.3.5.4	Documentation	35
C.3.6	Studies, Analyses, and Evaluations.....	35
C.3.7	Administrative Support.....	35
C.4	CONTRACT ADMINISTRATION	Error! Bookmark not defined.
C.4.1	Program Management Office.....	Error! Bookmark not defined.
C.4.2	Task Order Management.....	Error! Bookmark not defined.
C.4.3	Meeting Attendance	Error! Bookmark not defined.
C.4.4	Status Reporting.....	Error! Bookmark not defined.

C.1 INTRODUCTION

C.1.1 Objective

The objective of the Enterprise Programs, Infrastructure, and Communications Services (EPICS) contracts is to obtain support services that will help programs within the Communications, Information and Network Programs Group (CINP) to accomplish their mission objectives. Each contract provides a broad range of System Engineering; Acquisition and Program Management Support; In-Service Management and Support; Business and Financial Management; Information Systems Development and Support; Studies, Analyses, and Evaluations; and other services required by the Group. The required capabilities and services include knowledge transfer of the latest technologies, methods, approaches, and products that can lead to new, innovative solutions and best practices within the organization. The contracts are expected to serve as the primary means for CINP to obtain required technical services.

C.1.2 Background

The CINP is responsible for the implementation and execution of communications infrastructure programs supporting the National Airspace System (NAS) and non-NAS systems and networks. Two of the NAS programs—System-Wide Information Management (SWIM) and NAS Voice System (NVS)—are designated as “NextGen-enabling” programs because of their fundamental role in supporting the transformation of the NAS to the Next Generation Air Transportation System (NextGen). The EPICS Contractors provides vital support to the planning, procurement, implementation, and execution of these and other FAA programs as they progress through their lifecycles.

C.1.3 Scope

This Statement of Work (SOW) is comprised of distinct functional work areas that establish the scope of the contract. Specific requirements within the scope of these functional work areas will be identified, defined, and ordered by the CO via task orders. The Contractor must provide the full range of support capabilities defined in the task orders issued under this contract. The Contractor must furnish the necessary personnel, materials, equipment, and facilities required to satisfy the requirements of the individual task orders.

There are currently six groups and programs within CINP:

- Enterprise Programs,
- Enterprise Infrastructure Services,
- Air/Ground Voice Communications,
- NAS Voice Systems,
- SWIM, and
- FAA Telecommunications Infrastructure (FTI-2).

The scope of the contract includes support of all CINP and Data Communications programs and initiatives, as required, consisting of but not limited to the following:

Group or Program	Systems and Networks
Enterprise Programs, Enterprise Infrastructure Services, and FTI-2	• <i>FTI – FAA Telecommunications Infrastructure</i> • <i>FTI-2</i> • <i>ACS – Aeronautical Communications Services</i> • <i>ADAS – Advanced Weather Observing System (AWOS) Data Acquisition System</i> • <i>ASTI – Alaskan Satellite Telecommunications Infrastructure</i> • <i>BWM – Bandwidth Manager</i> • <i>DMN – Data Multiplexing Network</i> • <i>FAVES – FAA Administrative Voice Enterprise Services</i> • <i>FOMS – FAA-Owned Microwave Systems</i> • <i>IFMS – Invoice and Financial Management Services</i> • <i>Networx</i> • <i>MEVA – Mejoras al Enlace de Voz del ATS</i> • <i>NADIN PSN – National Airspace Data Interchange Network Packet Switch Network</i> • <i>NMR – NADIN Message Switch Network (MSN) Replacement</i> • <i>WMSCR – Weather Message Switching Center Replacement</i>
Air/Ground Voice Communications	• <i>VHF A/G Comm</i> • <i>Next Generation Air/Ground Communications (NEXCOM) Segment 2</i> • <i>Communications Facilities Enhancement Program</i> • <i>Radio Control Equipment</i> • <i>Radio Frequency Interference</i> • <i>UHF Radio Replacement</i> • <i>Airport Cable Loop Program</i>
NAS Voice Systems	• <i>NVS – NAS Voice System</i> • <i>VSRS – Voice Switching and Recording Systems</i> • <i>TVSR II – Terminal Voice Switch Replacement Program II</i> - <i>IVSR – Interim Voice Switch Replacement</i> - <i>VSBP – Voice Switch Bypass</i> - <i>ETVS – Enhanced Terminal Voice Switch</i> - <i>RDVS – Rapid Deployment Voice Switch</i> - <i>STVS – Small Tower Voice Switch</i> - <i>CCS – Conference Control Systems</i> • <i>NGVRRP – Next Generation Voice Recorder Replacement Program</i>

Group or Program	Systems and Networks
	• VSCS – Voice Switching Control System
System-Wide Information Management	• SWIM – System-Wide Information Management
Data Communications	• DataComm – Data Communications • DCIS – Data Communications Integration Services

This information is provided for information purposes and is subject to change. The list of programs and initiatives is expected to evolve during the EPICS period of performance. The Contracting Officer (CO) may also order support services for non-CINP-managed FAA programs and organizations.

C.2 GENERAL

C.2.1 Terminology

Within this SOW and associated task orders, references to “the Contractor” apply to the EPICS Contractor. References to “vendors” and “service providers” apply to other contractors from which CINP obtains systems and services.

C.2.2 Applicable Documents

Applicable FAA, DOT, and other Governmental orders and standards and commercial standards will be specified within the individual task orders. Unless otherwise noted, the most recent version of each order and standard applies.

C.2.3 Government-Furnished Property (GFP)

The Government may provide Government-furnished property (GFP) when the Government determines it to be required for the Contractor to perform specific work efforts authorized under this contract and specifies the GFP to be provided in a task order. All GFP is the property of the Government and must not be transferred to any individual or agency, public or private, without the express written approval of the CO.

C.2.3.1 Wireless Internet Access and Remote Access to the FAA Intranet

When required to perform the specific duties and roles assigned under this contract, the Government may provide, at no expense to the Contractor, mobile broadband modems and remote access to the FAA Intranet to CO-authorized Contractor personnel. Government approvals of mobile broadband modems and FRAC accounts for remote access will be reassessed annually on an individual basis.

C.2.3.2 Workspace, Telephones, and Computers at Governmental Sites

For Contractor personnel working under a Government Site Rate as defined in Section B, the Government will provide workspace, a computer, software, a telephone, and access to network printers and fax machines for the performance of assigned work efforts under this contract. Contractor personnel must comply with the Government's applicable usage policies for all GFP.

C.2.3.3 Other GFP

Any GFP not described in clause C.2.3.1 or C.2.3.2 GFP, including information, material, and equipment, must be specified within individual task orders as required. Unless otherwise approved, the Contractor will not be provided with Government-furnished laptops or mobile phones.

C.2.4 Security Requirements

Additional security requirements, applicable to the specific work to be performed under each task order, may be identified within individual task orders.

C.2.5 Data Deliverables

Specific technical-data deliverables will be included in individual task orders issued under this contract.

C.2.6 Quality Management

The Contractor is responsible for ensuring the quality of all work efforts and deliverables, cost control, and adherence to schedules. The Contractor must maintain a Quality Management System that is current to the latest ISO-9001 version.

C.3 FUNCTIONAL TASK AREAS

Seven top-level functional task areas are expected to be addressed within task orders issued under the EPICS contract:

- System Engineering;
- Acquisition and Program Management Support;
- In-Service Management and Support;
- Business and Financial Management;
- Information Systems Development and Support;
- Studies, Analyses, and Evaluations; and
- Administrative Support.

Within each of the top-level functional task areas are associated subcategories of activities as indicated in the following sections. An individual task order may address one or more functional task areas or subcategories. The following sections describe the work that must be completed under each task area if ordered; the Contractor must not perform any work described in section C.3 or C.4 unless specifically ordered via task order.

C.3.1 System Engineering

System Engineering is focused on the planning and development of new capabilities and improvements to existing capabilities. System Engineering involves, but is not limited to, identifying and analyzing requirements, assessing system designs and architectures, and planning and evaluating test activities to help to ensure that vendor-provided solutions meet the FAA's operational needs. To perform the required work efforts in this area, the Contractor must possess, maintain, and apply an in-depth understanding of industry-standard reference architectures, the NAS Enterprise Architecture framework, and NAS subsystem functions and interfaces.

C.3.1.1 Requirements Definition and Analysis

The Contractor must support the definition and analysis of functional and performance requirements. At a minimum the Contractor must

- Perform requirements-management activities to identify and manage the requirements that describe the desired characteristics of a system;
- Define, collect, document, and manage requirements for Initial Requirements Documents (IRDs), Final Requirements Documents (FRDs), Service Specification Documents, Interface Requirements Documents (IRDs), Interface Control Documents (ICDs), and other system and procurement specifications;

- Perform lifecycle engineering to identify and manage requirements for system lifecycle attributes including but not limited to deployment and transition, integrated logistics support, sustainment and technology evolution, and disposal;
- Perform functional and system analyses to translate stakeholders' needs into a traceable set of functional and system requirements;
- Analyze and define requirements for capabilities to support NAS-wide information sharing including but not limited to repositories, registries, Extensible Markup Language (XML) gateways, policy servers, and message-oriented middleware;
- Collect, analyze, and track service requirements submitted by service users;
- Maintain near- and long-term projections of service demands to facilitate network and system capacity planning; and
- Assess and project when technology enhancements may be required to satisfy user requirements.

C.3.1.2 Concept Analysis

The Contractor must provide support associated with identifying and analyzing candidate alternatives and concepts that address system issues, shortfalls, technological defects, end-of-service issues, and other issues. At a minimum the Contractor must

- Ensure compatibility with NAS architecture;
- Support lifecycle cost estimation and cost-benefit analysis;
- Perform risk assessments;
- Evaluate supportability;
- Evaluate operational and procedural impacts;
- Evaluate environmental impacts; and
- Evaluate overall NAS schedule impacts.

C.3.1.3 Enterprise Architecture Planning and Analysis

The Contractor must support the development of the NAS Enterprise Architecture (EA) by supporting the proper incorporation of supported programs into NextGen planning. The Contractor must support the development and refinement of artifacts that describe the evolution from the current NAS EA to the NextGen EA. At a minimum the Contractor must

- *Support strategic mission analysis, program mission analysis, and investment analysis; develop operational concepts for supported programs and their translation into NextGen capabilities, operational improvements, and performance requirements;*
- *Develop and assess alternative architectural frameworks (e.g. service-oriented and component-based) for system interfaces and information exchanges across the NAS enterprise;*

- *Perform Enterprise Infrastructure Assessment of CINP and customer programs at the various phases of the AMS acquisition lifecycle;*
- *Support the development and assessment of Enterprise Information Management (EIM) plans and enterprise-level and system-level data architectures;*
- *Support the application of the FAA's selected architectural framework for NextGen and NextGen-enabling systems; develop reference models, address associated security issues and challenges, and possess and maintain a working knowledge of associated development products, technologies, platforms, standards, data formats, and protocols;*
- *Perform operational and development analyses and studies for the development and management of the NextGen EA;*
- *Track emerging technologies in the commercial marketplace and provide recommendations to the FAA about potential opportunities to improve performance and reduce cost;*
- *Support the development of technology-evolution plans and transition strategies for supported programs as applied to the NextGen EA; and*
- *Identify and assess supported system alternatives and concepts to address IT-architecture gaps, capability shortfalls, technological defects, risks, and other issues.*

C.3.1.4 Standards Development, Validation, and Coordination

The Contractor must support development of international standards for aviation safety, performance, and interoperability. At a minimum the Contractor must

- Support standards-development organizations by acting as subcommittee coordinators, providing meeting minutes, collecting and distributing papers and materials, maintaining membership lists, and serving as administrative focal points for working-group activities;
- Support the delegation lead, committee, and working-group chair by preparing standards material, editing and integrating contributor submissions, collecting and resolving reviewer comments, and coordinating with other standards bodies and air navigation service providers;
- Provide standards expertise to support other development activities via technical-interchange meetings, teleconferences, and reviews;
- Support domestic and international user forums and panels to coordinate implementation plans and activities; and
- Support prototype development and validation efforts being conducted at FAA test facilities, vendor sites, and designated field locations.

C.3.1.5 Program- and Project-level Engineering

The Contractor must support Program- and Project-level Engineering activities. Program- and Project-level Engineering consists of activities specific to supported programs and projects with a particular emphasis on satisfying stakeholder requirements. At the beginning of a program or project lifecycle, the conceptual Enterprise Architecture is used as an input to develop a detailed architecture, system design, and implementation plan. All these elements must be developed in coordination and collaboration with stakeholders and other support contractors to achieve organizational goals. After implementation, Program and Project Engineering continues, and the focus shifts toward enhancements and optimization.

At a minimum the Contractor must

- Develop, propose, and maintain System Engineering Management Plans;
- Collect and validate requirements, provide architectural trade-offs, develop cost estimates and funding requirements, maintain configuration management of the solutions, provide telecommunications provisioning service, and track program and project implementation;
- Develop, propose, and maintain the technical-specification requirements;
- Perform radio frequency (RF) engineering and spectrum-management-related activities to support the development and deployment of air/ground communications systems and components;
- Support the definition and conceptualization of design prototypes;
- Apply Service-Oriented Architecture (SOA) design principles to develop SOA-based architectures to facilitate the organization and utilization of distributed capabilities that may be under the control of different ownership domains;
- Perform Enterprise Infrastructure Assessments encompassing all aspects of the infrastructure, e.g. networking, system, cloud computing, and SOA;
- Possess, maintain, and apply expertise in cloud computing and the potential impacts on telecommunications service needs;
- Develop system architectures and supporting elements including but not limited to conceptual data models, data descriptions, physical schemas, taxonomies, core service definitions, logical data models, XML schemas, and use cases;
- Establish, maintain, and update a repository or registry of system-provided core services;
- Possess, maintain, and apply an in-depth understanding of software-engineering practices, software-development standards, and middleware technologies to support oversight of the development and implementation of systems;
- Perform reliability, maintainability, and availability (RMA) analyses;
- Advise whether proposed designs comply with FAA and other policy, orders, and standards;

- Identify and assess the impacts of proposed technology upgrades;
- Perform engineering trade studies;
- Develop and review Program Control Board (PCB) items;
- Review NAS Change Proposals (NCPs) and provide recommendations for approval;
- Review planned contract modifications that have potential technical, performance, or cost impacts;
- Review and assess vendor requests for requirements deviations and waivers;
- Perform spectrum analyses to support the planning and implementation of satellite, microwave, and wireless-based communications systems;
- Identify, propose, and document telecommunications-service requirements (e.g. data rate, latency, and availability); advise whether telecommunications requirements submitted by customers comply with FAA and other policy, orders, and standards; develop, propose, and maintain in a repository all Customer Agreements;
- Coordinate the provisioning of telecommunications services via telecommunications architecture, requirements change board, and schedule change board reviews, and track and provide feedback on telecommunications-service requirements through implementation;
- Possess, maintain, and apply an in-depth understanding of the Technical Operations Safety Management System;
- Support the development of system-safety plans; review vendor-generated safety documentation; conduct and document system-safety assessments and hazards analyses;
- Provide technical assistance in identifying and resolving **Environmental, Occupational, Safety, and Health (EOSH) issues to ensure compliance with applicable laws, regulations, and policies;**
- Support planning for future systems, networks, and services;
- Interface with stakeholders that include but may not be limited to FAA programs, aviation-industry partners (domestic and international), aviation-regulatory agencies such as ICAO, and military organizations; and
- Work with stakeholders to understand and document their requirements, system architectures, security requirements, and data flows.

C.3.1.6 System Development and Integration

The Contractor must support system development and integration activities.

At a minimum the Contractor must

- Support the development of requirements, specifications, and requirements traceability;

- Review system documentation for the design, checkout, integration, and testing of systems at factories, test sites, and field sites;
- Monitor manufacturers' compliance with specifications;
- Perform technical analyses related to systems-hardware engineering, system interfaces, software engineering, and software maintenance;
- Support the design and implementation of interfaces among systems;
- Perform technical evaluations and analyses of CINP systems in the areas of test planning, safety, logistics, and service-life-extension planning;
- Support technical-interchange meetings and design reviews on CINP system components, software, and interfaces; this includes reviewing design changes and advising whether the system meets all requirements of specifications and referenced standards as well as required and standard engineering practices;
- Support working-group meetings concerning software, interfaces, hardware, and specialty engineering;
- Review upgrades and modifications of any non-developmental items (NDI) and identify any issues or concerns as well as present potential resolutions; and
- Provide technical assistance in documenting system-transition requirements, and assist in the generation of transition plans and in the actual transition of services.

C.3.1.7 Test and Evaluation Support

The Contractor must support the test and evaluation of systems. At a minimum the Contractor must

- Develop and review test plans and procedures;
- Support planning and coordination of test activities;
- Observe vendor-conducted testing and provide feedback to the FAA;
- Review vendor-developed test reports and provide recommendations to the FAA; and
- Execute test activities for functional, system-level, and operational testing.

C.3.1.8 System Production

The Contractor must support system-production activities. At a minimum the Contractor must

- Review system documentation for first-article testing at factories and production testing at one, some, or all test sites and field sites;
- Monitor manufacturers' compliance with production specifications;
- Observe first-article and production testing; and

- Perform technical analyses related to the production of systems, including problem management, hardware maintenance, system interfaces, and software maintenance.

C.3.1.9 System Performance

The Contractor must support system-performance activities. At a minimum the Contractor must

- Develop performance specifications;
- Monitor manufacturers' compliance with specifications;
- Collect, analyze, and evaluate system-performance data for new and existing systems;
- Review modifications to systems and analyze their impacts on system performance; and
- Support technical-interchange meetings, design reviews, and working groups.

C.3.1.10 Interface Management

The Contractor must provide system-engineering support in the area of interface management, which includes identification, definition, and control of interfaces. At a minimum the Contractor must

- Analyze and document whether pieces of systems work together to achieve the systems' goals and continue to operate together as changes are made during systems' lifecycles;
- Support the design and implementation of interfaces among new systems;
- Propose and illustrate performance, physical, and functional characteristics in sufficient detail for all details on the interfaces to be determined solely from the information in the Interface Requirements Documents (IRDs) or Interface Control Documents (ICDs);
- Propose required interface data and monitor submission of the data;
- Monitor the interface requirements and design to detect and report any changes to characteristics that might affect compatibility among systems and equipment; and
- Communicate coordinated interface requirements and design decisions and changes to program office participants.

C.3.1.11 Information Security

For both NAS and non-NAS systems, the Contractor must support the identification of threats; the assessment of risk and vulnerability; risk mitigation; incident detection and remediation; testing of proposed and applicable security solutions; and development of security policies and requirements. In addition, the Contractor must assist with evaluating, reviewing, defining, and providing recommendations regarding defense-in-depth security architectures; assessing the impact of proposed technologies; developing security plans; analyzing system and network security requirements; and validating systems-level solutions and alternatives. The Contractor

must have a working knowledge of all aspects of physical and personnel security and their interrelations with information security.

At a minimum the Contractor must

- Support all activities associated with establishing and maintaining security policies and procedures for systems and networks; this includes supporting the analysis and prioritization of appropriate security-measure enhancements and upgrades and verifying and analyzing security features incorporated into new and modified systems; this also includes administering the security of software development and distribution platforms and, for the physical security of software-development systems, analyzing the source and impact of incursions or attempted incursions and proposing and implementing corrective responses;
- Support all activities associated with establishing and maintaining security policies and procedures for NAS and non-NAS systems and subsystems related to the development, maintenance, and support of the FAA Enterprise Architecture Framework including but not necessarily limited to Information System Security (ISS) portions; this includes but may not be limited to the development of all required ISS NAS EA views, datasets, and databases; the secure storage of the same; and participation in FAA NAS EA working groups;
- Assist in the coordination, development, and enhancement of cybersecurity operations among solution providers and other FAA programs and organizations;
- Continually maintain its awareness of applicable FAA, NIST, OMB, and other appropriate orders, policies, and guidelines, and monitor programs and contracts to advise whether security measures are in compliance;
- Support the assessment of FAA security policies to advise whether they remain current as the technologies and mechanisms employed for sharing NAS data evolve;
- Provide information-security expertise to assist with analysis and certification and accreditation documentation for compliance with FAA, NIST, and other standards for accreditation; and
- Coordinate security audits and assist with the implementation of both short- and long-term mitigation strategies and alternatives for systems and networks.

C.3.1.12 Governance

The Contractor must develop, propose, and maintain governance strategies, policies, and plans for controlling system access and data distribution for supported programs and at a minimum must provide:

- Information standards and technologies including Web Service Risk Management, Security, Addressing, and Notification;
- Enterprise management and core-services development such as Interface Management, Messaging, Security, and Enterprise Service Management (ESM);
- Web Service Interoperability (WS-I) profiles including WS-I Basic, WS-I Basic Security, and WS-I Reliable Secure; and
- SOA security including XML Gateway, Lightweight Directory Access Protocol (LDAP), and Security Assertion Markup Language (SAML) and other industry-standard COTS solutions.

At a minimum the Contractor must

- Propose defined roles and responsibilities of governing bodies and stakeholders;
- Perform value analyses and propose service-selection criteria; and
- Audit, assess, and validate compliance with governance policies, processes, procedures, and requirements including but not limited to service-design standards, security policies, publishing and discovering standards, and the run-time enforcement framework.

C.3.1.13 Flight Procedures

At a minimum the Contractor must

- Provide flight-procedures and technical expertise and project-implementation and transition support for the Terminal Instrument Procedures (TERPS) Development program;
- Support obstruction-evaluation engineering and development of other applicable flight procedures; and
- Provide technical instruction on a variety of FAA orders and directives related to flight procedures, airspace, and TERPS.

C.3.1.14 Human Factors

At a minimum the Contractor must provide human-factors engineering expertise and an in-depth understanding of the air traffic controller operating environment in both the terminal and en-route domains to support (1) the definition of operational requirements for CINP systems and (2) the integration of those systems into the air traffic controller operating environment, which includes ATC staff and supervisory positions, ATC automation, the traffic management unit (TMU), and flight services. The Contractor must support the refinement of existing ATC procedures and the development of new ATC procedures as new systems, tools, and capabilities are deployed to the air traffic controller operating environment.

C.3.2 Acquisition and Program-management Support

Acquisition and program-management support begins with assisting the FAA in acquiring the systems and infrastructure to meet both NAS and non-NAS requirements. Once a vendor has been selected, the Contractor must support the implementation and management of the programs throughout the remainder of their lifecycles from system development to implementation, operation, and decommissioning.

C.3.2.1 Acquisition Planning and Support

The Contractor must provide planning and support for acquisitions. The Contractor must possess and maintain an in-depth familiarity with the FAA Acquisition Management System (AMS) and the FAA Acquisition System Toolset (FAST) to support programs throughout their acquisition lifecycles. The scope of the efforts range from small projects to major initiatives that require approval by the Acquisition Strategy Review Board (ASRB), FAA Chief Financial Officer (CFO), and Joint Resources Council (JRC). Activities may apply to new systems and networks as well as existing systems and networks.

At a minimum the Contractor must

- Assist in conducting market surveys, preparing notifications of planned solicitations, and responding to vendors' inquiries;
- Provide expertise on the acquisition of commercial off-the-shelf (COTS)-based solutions and support their acquisition;
- Perform mission-shortfall analyses and draft justifications to obtain funding and approval to address shortfalls;
- Support the development of independent Government cost estimates (IGCEs) and business-case analyses (BCAs);
- Assist in preparing the documents and briefings required to obtain program and project approval including those items associated with AMS milestones; these documents may include but are not limited to Acquisition Program Baselines and execution plans, Concept and Requirements Definition documents, Investment Analysis Readiness Decision documents, Initial Investment Decision documents, Final Investment Decision documents, Implementation Strategy and Planning Documents, and Program Management Plans;
- Assist with the preparation of Screening Information Requests including the development of technical specifications, statements of work, and data-item descriptions for contract deliverables;
- Support source-selection evaluations;
- Support the development of Government counteroffer positions; and
- Assist with contract definitization and program start-up for new programs and follow-on efforts.

C.3.2.2 Program-management Support

The Contractor must support the management of programs. At a minimum the Contractor must

- Develop and maintain appropriate management, scheduling, and tracking systems to facilitate program management and oversight;
- Participate in program- and project-management discussions including program-management meetings and reviews;
- Review vendor deliverables and provide comments and recommendations on Governmental approval;
- Establish and maintain online repositories and reference libraries of program documentation, and provide access to authorized stakeholders;
- Develop and propose program metrics; support the monitoring, tracking, and reporting of program metrics;
- Assist with proposed contractual changes by identifying affected contractual language; assess the cost, schedule, and performance impacts of proposed contractual changes;
- Assist with the development, administration, and tracking of service-level agreements with service providers;
- Develop quality-assurance (QA) plans; review vendor-generated QA plans; and identify, document, and report on QA issues, problems, and concerns as they arise;
- Assist with the development and execution of interagency agreements (IAs) and memoranda of agreement (MOAs) with external stakeholders; and
- Assist with the tracking and administration of legacy and other telecommunications services obtained through third parties such as the Defense Information Systems Agency (DISA), the General Services Administration (GSA), and local exchange carriers (LECs); establish and maintain a database of all awarded accounts; identify accounts that are candidates to be discontinued or moved to other suppliers; establish, maintain, and apply proficiency with third-party ordering and invoicing systems including, but not limited to, DISA Direct and GSA ordering tools.

C.3.2.3 Outreach

The Contractor must support internal and external communications, coordination, and education activities pertaining to programs and initiatives. At a minimum the Contractor must

- Provide operational input for development of materials used for outreach efforts, including videos, promotional materials, participatory demonstrations and events, conferences, and other interactions with users and other constituents;
- Develop materials that describe planned system benefits to the aviation community and to the NAS;

- Support development of interactive ATC and flight-deck simulations to provide realistic demonstrations of future systems and concepts;
- Provide industry, airline, and user subject-matter expertise to facilitate interaction with industry and the user community and convey user feedback in a constructive and positive way; and
- Support user-forum activities and other airspace-constituency coordination activities.

C.3.2.4 Solution-development and Implementation Support

The Contractor must support the solution-development and implementation phases of programs. At a minimum the Contractor must

- Develop implementation and integration plans and schedules that make effective use of FAA and vendor resources and minimize disruptions to FAA operations;
- Maintain supplemental information, including but not necessarily limited to activity descriptions and resource-loading, associated-cost, and percent-complete data;
- Provide and review critical-path schedule networks, Program Evaluation and Review Technique (PERT) analyses, and Gantt charts;
- Support the application of industry best practices and tools to the oversight and management of vendor-development activities;
- Support design reviews, site inspections, and surveys;
- Define and maintain enterprise standards for integration artifacts (e.g. canonical message definitions and consistent Web services interface definitions);
- Maintain integration metadata including but not limited to adapter specifications, network business objects, transformation and routing rules, XML message schema, call signatures, file formats, screen layouts, and network business object definitions; and
- Observe vendor-conducted tests and review vendor-generated test reports.

C.3.2.5 Program Management Assessments and Reporting

The Contractor must perform assessments and compile data to support program-management reporting commitments and responses to programmatic inquiries. At a minimum the Contractor must

- Assist with preparation of weekly and monthly activity reports; compile the inputs from individual FAA program managers and functional leads into consolidated reports; edit and coordinate the compiled report with appropriate recipients;
- Support the preparation of OMB Exhibits 53 and 300 submissions;
- Support Earned Value Management (EVM);
- Support Integrated Baseline Reviews (IBRs);

- Assist with the definition of strategic goals, track actual performance relative to goals, and support FAA reporting on performance results;
- Support Post-Implementation Reviews (PIRs) and Service-Level Reviews (SLRs);
- Assist with completing In-Service Review (ISR) checklist items; coordinate with stakeholder organizations and ISR checklist-item owners; track status and completion of checklist items.
- Assist with developing risk-management plans; identifying, documenting, and assessing program risks; developing risk mitigation plans; and tracking risk mitigation efforts;
- Support management's reporting to the FAA Engineering Council, JRC, and Department of Transportation (DOT) Investment Review Board (IRB);
- Assist with responding to congressional, inspector-general, Government Accountability Office (GAO), and other external inquiries;
- Compile data, perform analyses and assessments, and prepare briefings and white papers to convey the data and explain how it should be interpreted; and
- Maintain a repository of responses to ensure consistency and support comparative analyses of changes over time.

C.3.2.6 Quality Assurance

The Contractor must provide systems-engineering support for quality-assurance activities. The Contractor must provide support to programs and to FAA Quality Reliability Officers (QROs) assigned to programs.

At a minimum the Contractor must

- Develop and propose quality requirements for new system acquisitions;
- Perform evaluations during site surveys and installations to assist with determining other contractors' adherence to quality requirements;
- Monitor other contractors' performance in the area of quality assurance; and
- Assist with determinations and documentation of whether to accept or reject acquired systems, equipment, and material.

C.3.2.7 Risks, Issues, and Opportunities (RIO) Management

The Contractor must provide systems-engineering support managing and tracking risks, issues, and opportunities (RIOs). At a minimum the Contractor must

- Identify systematic and programmatic RIO items;

- Classify system RIO items as they apply to technical, cost, or schedule impacts;
- Identify and analyze risk- and issue-mitigation strategies;
- Prepare risk assessments and mitigation strategies for the RIO Board;
- Document RIO items and track them to closure; and
- Provide periodic reports on RIO items, their impacts, and associated mitigation strategies.

C.3.2.8 Organizational Metrics

At a minimum the Contractor must

- Assist with the formulation, preparation, tracking, and reporting of organizational metrics; these metrics will determine the quantity and quality of services provided by programs in support of organizational and FAA goals;
- Develop and implement mechanisms for the collection, reporting, and dissemination of data needed for these metrics; and
- Provide reports on organizational metrics at scheduled intervals and on an ad-hoc basis.

C.3.2.9 Configuration Management

The Contractor must support configuration management by providing knowledge and expertise to ensure that functional and physical characteristics of system-configuration items meet the requirements during all phases of systems' lifecycles. The Contractor must review specifications, contracts, and statements of work to ensure requirements are properly and accurately documented in all configuration-management-related data requirements.

At a minimum the Contractor must

- Analyze and report each system's compliance with FAA Order 1800.66;
- Assess the accumulation of data configuration, cost, schedule, and technical and operational impacts of proposed changes;
- Review NAS Change Proposals (NCPs) for impacts on projects, develop change summaries, and provide recommendations;
- Review configuration changes, assess cost and schedule impacts of such changes, and recommend whether formal change proposals are required;
- Collate data for formal changes, generate case files in an FAA status system, and follow up on resolutions of outstanding comments; and
- Provide administrative and technical support appropriate for the changes under consideration by Configuration Control Boards (CCBs).

C.3.3 In-Service Management and Support

The Contractor must support In-Service Management and Support, which is focused on the operational phase of program lifecycles and involves planning and coordination, in-service management, sustainment, provisioning, performance assessment and reporting, operational improvements, field-operations support, and product and integration support.

C.3.3.1 Planning and Coordination

The Contractor must support the planning and coordination of activities that have potential impacts to the operation of systems and networks. At a minimum the Contractor must

- Develop and propose provisioning and integration plans that meet the needs of all stakeholder organizations;
- Coordinate with affected facilities, organizations, vendors, and service providers regarding system and service transitions;
- Develop and propose implementation schedules for system improvements and technical refreshes that minimize disruptions to FAA operations and loss of service;
- Track all planned implementation activities related to new systems and services as well as changes, upgrades, and disconnects of existing systems and services; this includes reporting whether prerequisite actions are scheduled and completed appropriately; prerequisite actions include but are not necessarily limited to
 - Preparing FAA and vendor sites;
 - Coordinating site access;
 - Scheduling maintenance releases; and
 - Delivering required infrastructure (e.g. telecommunications connectivity, customer premise equipment, and breaker panels);
- Assist with developing change-management processes, conducting change-impact assessments, and validating changes to operational services;
- Track and assist with the coordination of the cutover of services to operation as soon as practicable after service deliveries and changes are accepted by the FAA;
- Support the development of configuration-management plans and review and report on vendor-generated configuration-management plans;
- Develop, propose, and maintain COTS product-management plans;
- Develop, propose, and maintain maintenance concept-of-operations plans;
- Track change requests related to new requirements and system enhancements, changes, and defects; perform technical assessments of requested changes, coordinate Program Control Board (PCB) meetings, assist with disposition of change requests, and document the outcomes of PCB meetings;

- Update configuration-management documentation and databases to reflect service deliveries, changes, and disconnects; and
- Assist with the development and conduct of Air Traffic Organization training.

The Contractor must support the development of the concepts of operations for systems, networks, projects, and initiatives. At a minimum the Contractor must

- Assist with the planning, development, and execution of Integrated Logistics Support (ILS) programs;
- Review and provide comments on vendor-developed maintenance and operations plans;
- Compile service-performance requirements from contracts, design documents, specifications, and standards;
- Develop and propose key performance metrics and evaluations of current service performance;
- Assist with the development, review, and update of FAA policies and orders;
- Provide technical assessment and feedback on potential operational impacts associated with vendor proposals and requests; and
- Collaborate with FAA program offices, prime contractors, subcontractors, and other stakeholders to establish SLAs that define commitments to response times, availability, data integrity, message volumes, and other measures.

C.3.3.2 In-Service Management

The Contractor must support the in-service management of systems and networks. At a minimum the Contractor must

- Support the management and monitoring of infrastructure in conjunction with FAA personnel, network managers, and system administrators by utilizing FAA- and vendor-provided monitoring tools to maintain visibility into operational statuses;
- Track the restoration of services when outages occur and support the tuning and troubleshooting of systems by reviewing vendor-generated trouble tickets to isolate faults within applications, networks, and integration infrastructure;
- Assess chronic operational issues and support collaboration with vendors on corrective-action plans;
- Review, assess, and provide feedback on vendor-generated trouble tickets and post-mortem reports, including the following:
 - Assess and validate the portion of restoration time attributable to the FAA and to other parties;
 - Identify opportunities to improve FAA and vendor practices; and

- Coordinate the follow-up activities, such as surveillance-evaluation records (SERs) and post-outage assessments, for significant outages;
- Assist with the resolution of issues with vendor field personnel;
- Support operations projects at the FAA Operations Control Centers (OCCs) and Network Enterprise Management Centers (NEMCs);
- Assist with the collection and tracking of Logistics Management Information (LMI), including spare-parts inventories and consumption rates;
- Support the planning and oversight of the decommissioning and disposal of systems and network infrastructure; and
- Conduct telecommunications site surveys and site-preparation work.

C.3.3.3 Sustainment

The Contractor must support sustainment-engineering activities for systems. At a minimum the Contractor must

- Identify system components reaching end-of-life or end-of-service dates;
- Identify and quantify the projected need for system components required to sustain current systems;
- Identify and quantify projected shortfalls in system-component availability or service life;
- Identify technological opportunities that may provide alternatives for sustainment engineering; and
- Support technical-interchange meetings, design reviews, and working groups.

C.3.3.4 Logistics Support

The Contractor must provide logistics support for systems. At a minimum the Contractor must

- Support the development of maintenance-concept and support strategies;
- Identify and report system components for which spares will be required, down to the Lowest Replaceable Unit (LRU);
- Identify what quantities of spares will be on-site and off-site;
- Facilitate coordination with the FAA Logistics Center on sparing strategies for each system;
- Identify and report components that may require lifetime buys due to end-of-life and end-of-service issues and propose sparing strategies for those components; and

- Track system outages, time to restore service, and spare usage to assess and report the effectiveness of sparing strategy.

C.3.3.5 Performance Assessment and Reporting

The Contractor must support performance-assessment and reporting requirements for systems and networks. At a minimum the Contractor must

- *Support the planning, development, and evaluation of FAA performance-measurement activities (both facility-level and systemwide);*
- *Advise whether metrics are aligned to DOT and FAA measures and goals, including NextGen goals;*
- *Support activities associated with the selection of Critical-to-Quality (CTQ) measures, identification of Key Performance Indicators (KPIs), and assessment of the system and process stability and capability of services;*
- Develop, propose, and track SLAs with prime contractors and service providers;
- Develop, propose, and apply methodologies to analyze network performance, including trend analysis and data-quality assessment, in areas including but not necessarily limited to vendor SLAs, customer time, and FAA outages; evaluate and report performance trends to detect performance degradation and assist with the identification of root causes of degraded performance, including potential integration or systemic issues;
- Collect service-performance and -usage data and evaluate and report the impacts of service usage and growth, project future performance shortfalls, and develop and propose implementation plans;
- Review, analyze, and provide feedback on vendor-submitted documents summarizing network performance and utilization;
- Perform impact assessments and report results;
- Provide reports on the performance status of systems and networks; and
- Assess and report existing or potential performance shortfalls and support the coordination with stakeholders to determine improvements or contractual changes needed in order to meet current and future service-performance requirements.

C.3.3.6 Operational Improvements and Field-Operations Support

The Contractor must support operational-improvement projects and coordinate the field-level activities associated with the deployment of new systems and services, facility relocations and consolidations, and system and service moves and changes. At a minimum the Contractor must

- Assist the FAA with planning the required work efforts, developing a completion schedule, and coordinating and tracking the execution of the required work efforts;

- Provide site-visit, survey, service-delivery, and cutover support, which requires thorough on-site coordination and documentation to ensure accuracy and completion of work assignments;
- Support the planning and coordination of site preparation, system and service implementation, and cutover activities;
- Develop and propose transition and implementation plans, working with FAA stakeholder organizations, vendors, and service providers to establish proposed responsibilities and timelines and provide general project-implementation coordination to support the efficient delivery of services for key programs; and
- Support NAS Plan Program reviews, documentation, master scheduling of all forecasted efforts, individual project scheduling to the task level, and requirements reviews and analyses; participate in cost and schedule reviews; and coordinate site cutover activities for key implementation efforts.

C.3.3.7 Product and Integration Support

The Contractor must provide product and integration support regarding interoperability between legacy systems and newly deployed capabilities. In addition, the Contractor must support the operation and maintenance of FAA-operated systems to assist with ensuring they continue to meet their performance requirements.

The Contractor must possess, maintain, and apply an in-depth understanding of the affected NAS systems and development-environment tools including but not limited to programming languages (e.g. C++ and JAVA); operating systems (e.g. UNIX, Linux, and AIX)' middleware technologies and systems (e.g. Java Messaging Service (JMS), Active Message Queuing (MQ), and IBM MQ); enterprise application platforms (e.g. .NET and Java 2 Platform Enterprise Edition); SOA products (e.g. IBM WebSphere, Oracle Application Server, WebLogic, and Fusion Middleware); and SOA platforms and standards (e.g. Universal Description Discover Integration and electronic business XML).

At a minimum the Contractor must

- Provide technical assistance to ensuring the successful integration of new and upgraded FAA systems and networks into the NAS;
- Assist with documenting and maintaining system Interface Requirement Documents (IRDs), NAS Change Proposals (NCPs), and new product evaluations, and assist with security risk mitigation;
- Provide guidance to project-design teams on the development of adapters to support system interfaces;
- Develop, propose, and maintain configuration plans, standard operating procedures, and policies; support the implementation of configuration-management practices; and

support the establishment and maintenance of required configuration-management liaison relationships;

- Assist with recording the status of configuration documentation and tracking revisions to assist with ensuring documentation is complete and consistent;
- Participate in product-audit planning, conduct audits, and produce audit reports;
- Track remaining service life of FAA systems, and advise the FAA when supportability issues arise; and
- Support the workforce-development-planning activities, management and tracking of training, and training analysis for FAA systems and support overall training development, including the development and review of training development plans, training-initiative requests, training-course design guides, instructor guides, course materials, and other documents as required.

C.3.4 Business and Financial-Management Support

The functional area of Business and Financial-Management Support focuses upon assisting with ensuring that CINP programs are fully funded and that budgets are properly executed and tracked. The Contractor must provide a range of planning, tracking, program-analysis, financial-analysis, and cost-estimating activities.

C.3.4.1 Budget Planning and Formulation

The Contractor must support the planning and formulation of program budgets. At a minimum the Contractor must

- *Develop and propose current-year and out-year F&E and O&M budget forecasts and justification documentation;*
- Assist with responding to budget inquiries, performing program budget and requirements analyses, developing impact statements, and participating in and conducting budget meetings including Capital Investment Team (CIT) reviews;
- Maintain a budget history for prior years and track the base level of funding for programs and projects;
- Formulate draft spend plans for organizational elements, and assist in the development of obligation plans by project and functional element;
- Coordinate internal content collection, conduct research and analysis, provide and consolidate program narratives and projected budget inputs, assist with ensuring consistency with other budget inputs and processes, coordinate and consolidate draft materials, and support budget reviews; and
- Support program inputs to FAA financial-management processes and tools including but not limited to Resource Planning Documents (RPDs), the Corporate Work Plan (CWP),

the DELPHI Funds Control Module, the Financial Management System (FMS), REGIS, and PRISM.

C.3.4.2 Budget Execution and Tracking

The Contractor must assist with budget-execution and financial-management activities and ensuring that budget targets are met.

At a minimum the Contractor must

- Maintain automated financial-tracking systems for both F&E and O&M budget obligations and expenditures, including individual financial transactions;
- Track and provide summary reports to the FAA on funding allocations, obligations, expenditures, funds remaining, and projected shortfalls and surpluses for each functional area, Detailed Obligation Plan (DOP), and service area;
- Track actual contract costs, review invoices, and prepare payment-recommendation documents that map specific invoice charges to appropriate accounting codes;
- Track compliance with quarterly spend plans, develop estimate to complete (ETC) projections, forecast required funding profiles, and support end-of-Fiscal-Year closeout for programs;
- Track expenditures by DOP to the level of FAA Work Breakdown Structure (WBS) elements and monitor, track, reconcile, report, and provide updates on all funds received and expensed to the level of individual FAA accounting codes;
- Extract, analyze, and summarize data from DELPHI and vendor invoices;
- Reconcile DELPHI payment data relative to FAA invoice certifications and original funds-transfer memoranda;
- Coordinate and track funding memoranda, procurement requests (PRs), and budget authorizations (BAs) associated with the funding of services;
- Support the development, execution, and tracking of reimbursable agreements (RAs) with external service users;
- Coordinate Military Interdepartmental Procurement Requests (MIPRs) among the FAA Air Traffic Organization, CO, and CFO's office; coordinate new agreements with appropriate stakeholders and work with the Agreement Coordinator and the above organizations for MIPR processing in the FAA Reimbursable Tool; advise whether the appropriate line of business (LOB) overhead fees and General and Administrative (G&A) fees are calculated correctly; and
- Establish and maintain proficiency with FAA, vendor, and third-party invoicing and financial-management tools for tracking program costs and performing analyses to support FAA decisionmaking.

C.3.4.3 Investment and Business-Case Analyses

The Contractor must develop and propose business-case analyses and investment analyses to document the financial trade-offs associated with programs and projects. At a minimum the Contractor must

- *Perform cost-benefit analyses for programs and related initiatives including cost-and-benefit models and bases of estimates;*
- *Provide economic analyses to support successful completion of Investment Analysis requirements;*
- *Assess the accuracy, completeness, and reasonableness of externally developed cost-benefit analyses;*
- *Assess the risks associated with costs, benefits, schedules, technical performance, human factors, safety, and security;*
- *Conduct the iterative trade-off analyses among cost, schedule, performance, and benefits used to determine the most advantageous and reasonable solution for a core set of requirements; and*
- *Provide JRC-checklist support.*

C.3.4.4 Cost Estimating

The Contractor must develop cost estimates to support budget planning, programmatic decision-making, and other activities. The cost estimates may encompass, but may not necessarily be limited to, program-baseline development, budget and lifecycle cost estimates, feasibility studies, sensitivity analyses, contract-evaluation support, cost-performance monitoring and analysis, and special studies such as the development of fee-for-service models. At a minimum the Contractor must:

- *Provide cost estimates for programs;*
- *Provide cost inputs to lifecycle-cost profiles formatted according to the most recent version of the FAA Standard Work Breakdown Structure (WBS); and*
- *Prepare cost estimates for alternatives analyses and what-if analyses; all cost estimates produced by the Contractor must clearly identify all pertinent assumptions upon which the estimates are based.*

C.3.5 Information-Systems Development and Support

The functional area of Information-Systems Development and Support encompasses the full range of the information-systems lifecycle and applies to tools that facilitate program oversight functions as well as operational support functions for FAA platforms such as NADIN, WMSCR, and ADAS. The scope of the required functionality for the tools also includes supporting online collaboration and interaction within programs and with external stakeholders.

C.3.5.1 Enterprise Management Tools Development and Support

Enterprise Management Tools are a collection of internally developed information systems used to support management and oversight. The current set of tools includes the following:

- a) Telecommunications Enterprise Management Tool (TEMT),
- b) Budget Planning and Management System (BPMS),
- c) Service Implementation Tool (SIT),
- d) Quote Order Tracker (QOT), and
- e) Contract Letter Processing Tool (CLPT).

Development and support will not necessarily be limited to these tools. At a minimum the Contractor must

- Maintain existing tools, implement enhancements, and develop new tools;
- Conduct alternatives analyses, cost-benefit analyses, make-or-buy studies, and other assessments to recommend whether the software requirements can be best met through commercially available software tools or custom software development;
- Provide planning, requirements analysis and design, troubleshooting, integration, installation, operations, maintenance, training, documentation, and administrative services for the application-development efforts; servers; Internet, intranet, and extranet deployments; internal and external connectivity access; and enterprise-scale Internet-technology deployments; and
- Develop additional applications for deployment on the FAA's infrastructure.

All current applications require complete software-development lifecycle support. The Contractor must support, at a minimum, the following application- and website-development services: system analysis, requirements collection, design, development, training, data conversion, testing, implementation, documentation, maintenance, operations, administration, enhancements, modifications, and end-user support of all websites, web-enabled data-entry systems, standalone and client-server applications, and collaborative computing products. The Contractor must provide support in the following defined areas of information technology:

- Requirements Analysis and System Design: Assist Government personnel in conducting requirements analysis and system design for websites and applications (standalone, client-server, network-based, and web-enabled), including but not limited to proposing project plans with milestones, conceptual and detailed system designs, and documented system requirements;

- **Development Support:** Develop websites and applications (standalone, client-server, network based, and web-enabled) using industry-standard software and standards adopted by the FAA;
- **System Testing:** Develop and propose test plans and conduct structured testing in the following areas: functional, unit, system, interface, alpha, beta, and integration, including but not necessarily limited to preparing and submitting test-analysis reports and coordinating the correction of all discrepancies found during the testing period prior to system acceptance and accreditation or as agreed with the Government;
- **Database Development and Maintenance:** Provide database-development and -maintenance support for all software-development-related activities, including but not necessarily limited to database-requirements collection and analysis, database design and modeling, database scripting, creating data marts and data warehouses, database backup and recovery, database performance and tuning, database upgrades and updates, and database consolidations;
- **System Implementation:** Perform implementation, operations, and lifecycle maintenance of all aspects of web-based systems and technology implementations including web-page development, database development, and support of coordination with agency groups and COTS vendors;
- **Performance Monitoring, Optimization and System Maintenance:** Conduct periodic performance measurement and evaluation activities that may lead to re-engineering and optimization of existing websites and applications to improve productivity, system performance, and network throughput; incorporate changing functional and technical requirements; and address any other constraints identified by the Government;
- **Configuration Management:** Review and develop CM artifacts and advise whether documents meet Capability Maturity Model Integration (CMMI) requirements; develop, propose, and refine testing procedures within the rational environment; maintain updates as changes occur within application code, databases, and operating environments; and maintain the systems required for configuration management and documentation support;
- **Knowledge Base:** Establish and maintain knowledge base for all software and support activities using established tools such as FAA's Knowledge Services Network (KSN); and
- **Special Requirements:** Perform specific tasks, which may occur less regularly, including but not necessarily limited to the following: provide assistance in planning and performing data conversions to Internet technologies; develop data conversion and validation routines; develop special applications as required; develop, propose, and maintain a continuous quality-improvement and -assurance program; document

requirements of existing and legacy applications currently in use; develop, propose, and manage a configuration-management program for all supported applications; and collect statistics to support sizing applications and improving performance.

C.3.5.2 Helpdesk Support

At a minimum the Contractor must:

- Provide helpdesk support to users of Contractor-developed and Contractor-maintained Enterprise Management Tools;
- Ensure helpdesk support is available from 8:00am to 5:00pm Eastern Time on Government workdays;
- Ensure the helpdesk support is accessible via, at a minimum, telephone, email, and a Contractor-developed website;
- Provide same-day response to inquiries received by noon on a Government workday and next-workday response to inquiries received after noon on a Government workday or anytime on a non-workday; and
- Coordinate with the National Service Center (NSC) when required to investigate and resolve issues.

C.3.5.3 Common Platform Support

The Contractor must provide systems and software engineering and programmatic support for the development of the Common Platform and its associated subsystems. The Contractor must assist with generating and implementing consolidated requirements at the end-system level. This functional area includes support of all aspects of the Common Platform initiative from development through sustainment.

At a minimum the Contractor must

- Develop and propose system architectures;
- Develop and propose functional system and software specifications;
- Develop software;
- Generate for review test plans, test procedures, and test reports and conduct unit and system tests;
- Generate for review system and software concept-of-operations documentation;
- Generate system and software users' manuals;
- Support software-configuration management;
- Validate and support the resolution of Problem Tracking Reports (PTRs);
- Prepare software releases;

- Provide system- and software-related field support;
- Develop and propose Interface Control Documents (ICDs);
- Perform database design and maintenance; and
- Generate for review security-related documentation for security Certification and Accreditation (C&A) purposes.

C.3.5.4 Documentation

The Contractor must recommend updates for and maintain all pertinent documentation for each functional area and facility including but not limited to the following: vendor COTS documentation; product specifications; site plans and topologies; system manuals; desktop application manuals; standard operating procedures; vendor application documentation; testing, quality-assurance, contingency, backup, recovery, and restart procedures; configuration-management-related documentation. Documentation must be made available in hard copy and electronic format.

C.3.6 Studies, Analyses, and Evaluations

The Contractor must provide organized, analytical assessments and evaluations of complex issues to support the improvement of programs, decisionmaking, management, and operations of the FAA, its activities, and its systems. At a minimum the Contractor must

- Prepare and conduct studies, reviews, and evaluations and provide recommendations on subjects of highly specialized areas of concern;
- Perform evaluations to enhance FAA understanding and develop alternate positions and solutions to various issues and problems;
- Provide recommendations on organizational policy and directives;
- Review and report on new policies and their effects on current programs, procedures, and operations; and
- Provide recommendations on emerging technologies and their application to enhance mission performance.

C.3.7 Administrative Support

At a minimum the Contractor must

- Support correspondence preparation, file maintenance, creation and production of documentation and presentation materials, technical editing, graphic services, and document assembly and binding;
- Assist in the development, preparation, and presentation of briefing packages, brochures, handouts, and other information materials in various mediums;

- Plan and schedule conferences and meetings, including administering background and support material, arranging facilities and coordinating accommodations, providing organizational and administrative support, and organizing post-meeting activities;
- Organize, prepare, and execute displays and exhibits for technical conferences and conduct project briefings; and
- Take minutes of all project meetings attended; the minutes must include (1) the date and purpose of the meeting; (2) the names, positions, organizations, phone numbers, and email addresses of all persons attending the meeting; and (3) a description of action items assigned, actionees, and due dates for the completion of the action items; the Contractor must submit the minutes to the cognizant FAA official for review and approval within 24 hours following the meeting; within 24 hours of receiving FAA comments, the Contractor must incorporate them all into the minutes and resubmit the minutes to the FAA official for distribution; the Contractor must not distribute meeting minutes unless directed by the FAA; the Contractor's participation in meetings must be approved orally or in writing by the FAA prior to attendance.
- Maintain internally and externally developed documents and briefing materials submitted to the FAA in the performance of this contract; these files must be organized; indexed by title, source, and date; and electronically accessible to the FAA through a Government-provided website such as KSN.